Compact Band Weighting Module Based on Attention-Driven for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) data have large numbers of bands that probably not all bands are equally informative and predictive for an effective HSI classification. Effective algorithms are highly desired in many real-world HSI applications, especially in cases requiring rapid learning with limited computing power. To address the abovementioned case, we present in this article a novel plug-and-play compact band weighting (CBW) module based on the attention-driven mechanism that evaluates different spectral bands according to their contributions to a given classification task. Compared to existing band weighting (BW) modules with tens of thousands of network parameters by deep learning, the proposed CBW is a lightweight module with only 20 parameters. Both model complexity and time cost are significantly reduced. The CBW module implements BW by making full use of the correlation among the adjacent spectral bands and spectral statistic information and, thereby, leads to the effect of recalibrated HSI. The experimental study has been conducted on three widely used HSI data sets, and results show the superiority of the proposed algorithm over current state-of-the-art methods of BW. The source code is available at https://github.com/JarvenYi/CBW.**

***Index Terms*— Attention mechanisms, band weighting (BW), deep learning (DL), hyperspectral image (HSI) classification, lightweight module.**

I. Introduction

THE past decades have seen wide applications of hyperspectral image (HSI) that includes urban mapping, forest monitoring, environmental management, and precision agriculture [1], [2]. HSI data are often acquired in the form of high-dimensional data cubes with hundreds of spectral bands of narrow wavelength width [3]. Compared with ordinary images, the main advantage of HSI lies in their high spectral dimensionality with hundreds of quasi-continuous spectral bands. The acquired pixelwise spectral signatures contain relevant information on a field-of-view (FOV) of interest and yield important cues for HSI classification tasks.

HSI data with large numbers of spectral bands often exhibit significant redundancy, leading to the "Hughes phenomenon" [4] and overfitting [5] in HSI classification problems. Therefore, methods for dimensionality reduction or band selection have been extensively studied and developed [6]–[8]. Although these methods are useful to improve classification performance, they inevitably lead to information loss of HSI data to various extents. Instead of band selection, the approach of band weighting (BW) treats spectral bands differently according to their contributions to the classification task. A support vector machine (SVM) with a feature weighting algorithm has been developed for HSI classification by feature comparison [9]. Alternatively, a band weighted SVM method has been investigated to perform linear classification with nonlinear kernels and iteratively optimize two convex subproblems with respect to both sample coefficients and band weights [10]. Yan *et al.* [11] proposed a BW method that maximizes the interclass distance to determine the band weight. It has also been analyzed to use spectral and spatial information for modeling the per-class band contribution [12]. However, the conventional BW methods lack the ability to automatically and iteratively correlate band information for significantly improved classification accuracy. Therefore, enhancing BW algorithms is highly desirable with the adaptive capability to extract effective features.

The attention mechanisms by the human brain have been adapted recently to extract critical information for achieving a design goal by neglecting information of less importance or relevance [13], [14]. Different models based on attention mechanisms have been implemented in deep neural networks for various applications [15]–[18]. Such attention-driven models have achieved significant success in natural language processing, which includes machine translation [19], problem-solving [20], and natural language reasoning [21]. For image data processing, mapping channel relationships by squeeze-and-excitation networks (SENets) has been proposed to adaptively recalibrate channelwise feature responses by explicitly modeling the degrees of interdependence among channels [18].

It can be noted that the methods of BW and attention mechanism share a common idea of amplifying useful information and attenuating nonessential ones. The attention mechanism

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can perform such functions adaptively through deep learning. It attracts significant attention in the field of HSI BW. For example, Wang *et al.* [14] proposed a BW network that can learn the relationship among the bands and recalibrate them according to their joint contribution to classification by attention mechanism. A prominent aspect of BW is that it can assign different weights to different pixel areas by pixel-level weighted structure. Despite the advantage, improvement to the BW network is highly desired to reduce its complexity and enhance efficiency. For this purpose, Mou and Zhu [22] proposed a function-agminated structural spectral attention module (specAtteNet) that makes use of a gating mechanism to adaptively recalibrate spectral bands by selectively emphasizing informative bands and suppressing those deemed less useful. While the spectral attention module-based convolutional neural networks (CNNs) offer improved results, the spectral gates are derived by global convolution. Compared with global average pooling, global convolution is useful but complicated computation to suppress the effect of noise and requires a massive number of parameters. A band attention module (BAM) has been introduced with the implementation of DL-based HSI classification to weight bands [23]. BAM consists of five 2-D convolutional layers, and related networks exhibit improved performance. Still, incorporating BAM increases markedly model complexity and computational expense. From the above discussion, one can clearly see that these existing approaches vary in network topology, but all share certain limitations.

1) They achieve better performance at the cost of higher model complexity, which renders further improvement and compact implementation difficult. The general trend has been to make more complicated BW modules in order to achieve better performance. However, the classification task needs to be carried out in a timely fashion on a limited computational device.
2) These methods capture dependencies across all spectral bands despite the information redundancy in certain bands. However, it may increase unnecessary parameters that may be inefficient. In HSI BW, aggregating multiply adjacent spectral bands correlation may contain more available information that requires fewer parameters.

To solve the abovementioned problems, we propose a compact band weighting (CBW) method in the form of a network module, which yields band weights efficiently at the cost of a slight increase in model complexity. The CBW module first utilizes a global average pooling for each spectral band to squeeze given HSI data into a band statistics vector. Each 1-D convolutional layer is then applied to capture correlations of different orders among adjacent bands that are combined to form a band statistics vector. Finally, HSI is recalibrated by a band weight vector generated by aggregating multiply correlation among adjacent bands.

The main contributions of our work are summarized as follows.

1) We develop a highly efficient plug-and-play CBW module to improve the accuracy of various DL-based HSI classifications. The CBW module requires much fewer adjustable parameters number than existing BW methods and yields higher performance for HSI classification.
2) A novel approach has been developed in the CBW module that performs recalibration of spectral band-based band weights by capturing and aggregating multiply correlations among adjacent bands. The combination of multiply adjacent bands correlations provide a mechanism to fully utilize spectral information for significant improvement of bands weighting of HSI data for classification.
3) Extensive evaluations of CBW have been performed with ablation studies on three widely used HSIs. Experimental results on limited computing devices demonstrate that the proposed CBW performs well in terms of effectiveness and performance.

The remainder of this article is organized as follows. Section II describes related works on CNN-based HSI data classification and attention models in deep learning. Section III presents the principle and design of our CBW module. Section IV provides the experimental results obtained with different data sets and compares them against those other methods. Section V concludes this article with some remarks and discusses possible future improvements.

II. Related Works

A. HSI Classification

In general, HSI classification problems are tackled in two ways: one with traditional feature extraction technique and another with DL-based automatic feature extraction technique.

Numerous kinds of classification approaches based on traditional feature extraction techniques have been developed for HSI classification. SVM is one of the most common traditional methods [24]–[28] for classification. Its basic model is a linear classifier with the largest interval defined in the feature space. On the basis of SVM, many scholars have designed a variety of HSI classification algorithms. Fang *et al.* [29] have utilized the local covariance matrix to encode the relationship between different spectral bands based on SVM. Wang *et al.* [30] designed a multiple kernel-based SVM classification by combining spectral, spatial, and semantic information. However, the SVM and methods based on SVM are characterized as shallow network models [31] compared to deep networks, which can extract multilevel and abundant semantic feature representations.

Over the last decade, DL has achieved unprecedented success in the field of HSI classification [32]. With a powerful ability to extract complex image features, different CNNs have been investigated to accomplish HSI classification [33]. In [34], a CNN-based spatial–spectral feature extraction framework is proposed to directly process the small cubes of HSIs. In [35], complementary and correlated information has been considered among different hierarchical layers to build a deep feature fusion network (DFFN) by residual learning, which fuses the outputs of different hierarchical layers. Cheng *et al.* [36] have designed off-the-shelf CNN models for HSI classification by the unified metric learning-based framework to alternately learn discriminative spectral–spatial features. Chen *et al.* [34] and Zhong *et al.* [37] have designed the deep feature extraction scheme of 3D-CNN

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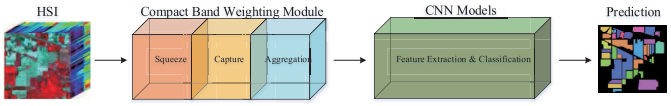


Fig. 1. Overall flowchart of CNN models based on CBW module for HSI classification. The CBW module consists of the squeeze part, the capture part, and the aggregation part. The CBW module can work any CNN model (take a single CNN as an example in this article).

and the spectral–spatial residual network (SSRN) by utilizing 3D-CNN. Roy *et al.* [38] combine 2D-CNN spatial feature learning layers and 3D-CNN spectral–spatial feature learning layers to establish HybridSN. These DL-based feature extraction algorithms show strong capability for classification tasks.

B. Attention Models

Attention models are inspired by the human brain's activity to efficiently utilize available resources when processing the most important information in the scene [18]. For DL-based analysis, an attention model is implemented as a module embedded into a network layer to generate a weighting mask. The mask is then applied to the feature maps to guide extraction toward those important features and dedicate fewer resources to irrelevant ones. In the area of computer vision, several efforts are now being made to add visual attention mechanisms into DL networks, improving the efficiency of the convolution kernels for equal treatment [39]. The SENet in [18] presents an effective, lightweight gating mechanism to self-recalibrate the feature map by means of channelwise importance [40]. In the remote sensing literature, several attention-driven techniques have been developed for the classification task. For example, Haut *et al.* [39] incorporate attention mechanisms to a ResNet in order to better characterize the spectral–spatial information contained in the data. Sun *et al.* [41] proposed a spectral–spatial attention module that is to suppress the effects of interfering pixels. Its attention module exploits the correlation between the hyperspectral pixels within an HSI cube to obtain attention areas and capture spectral–spatial attention features. Zhu *et al.* [42] proposed an end-to-end residual spectral–spatial attention network (RSSAN) for HSI classification, where a spectral–spatial attention module is designed for spectral BW and adaptive weighting of spatial information. These DL-based algorithms have improved the quality and efficiency of feature extraction by utilizing attention mechanisms.

III. Proposed Method

Our CBW is a plug-and-play module to be added into an existing CNN network, as shown in Fig. 1, which transforms an HSI patch I to map $\widetilde{I}$ of the same size by recalibrating the spectral bands. The architecture of the proposed CBW module consists of three components: squeeze, capture, and aggregation. The first component compresses the spatial information in the HSI data to be used for band representation by a vector of band statistics. The second one obtains two vectors of band dependence to quantify correlations of adjacent bands. The third component generates band weights by aggregating information of the band statistics and the correlation of adjacent bands for recalibrating HSI. Collectively, we can represent CBW as

$$\widetilde{I} = \mathrm{CBW}(I) = A(C(S(I))) \tag{1}$$

where $S(\cdot)$ describes the squeeze component, and $C(\cdot)$ and $A(\cdot)$ represent the interaction capture and aggregation component, respectively. The squeeze component is implemented as global average pooling to produce a vector of bandwise statistics vector v_s from the HSI patch I. Then, v_s is fed into the dependencies capture component to extract local cross-band correlations by 1-D convolution by two dependencies vectors of v_{d1} and v_{d2}. For the aggregation component, a band weight vector v_w is obtained by integrating the statistics and dependencies vectors to yield the recalibrated HSI patch.

As mentioned earlier, our CBW module consists of three components of squeeze, capture, and aggregation that are described in detail in the following, and the architecture of the CBW module is shown in Fig. 2.

A. CBW Module

To obtain the global information to each band in the initial HSI patch I, we squeeze HSI patch I into a vector of bandwise statistics vector v_s. The squeeze operation can be achieved by several schemes, global average pooling, global maxpooling, global 2-D convolution, and global 3-D convolution. For arriving at a compact approach to obtain global information, we have explored and eventually employed a global average pooling layer with no adjustable parameters to perform squeeze operation. The HSI patch I can be expressed as a stack of 2-D images of $I_1, I_2, \ldots, I_C$, with $I_c \in \mathbb{R}^{H\times W\times 1}$. The 2-D image of each band I_c is compressed into an average intensity by a global average pooling layer, so the cth element of the band statistics vector v_s can be expressed as

$$v_s^c = \mu(I_c) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} I_c(i,j), \quad c = 1, 2, \ldots, C \tag{2}$$

where $\mu(\cdot)$ is the global average pooling function, and $I_c(i, j)$ is the spectral intensity corresponding to the spatial location (i, j) of the image I_c.

Next, the capture component quantifies the nonlinear spectral band dependencies by their correlations. In HSI, adjacent spectral characteristics may correlate strongly, and these correlations play a significant role in accurately weighting the band characteristics. A fully connectional (FC) layer is the most common method to capture global correlation. However, the FC layer needs a large number of adjustable parameters and is, thus, inefficient to capture correlations among overall spectral bands, especially on extracting local dependencies from adjacent bands. We have utilized 1-D convolution operations to efficiently capture the correlation among adjacent bands. As shown in Fig. 3, the weight parameters of the 1-D convolution kernel can be used as factors to multiply the K adjacent band features for obtaining a feature, which implies the nonlinear correlation between adjacent bands.

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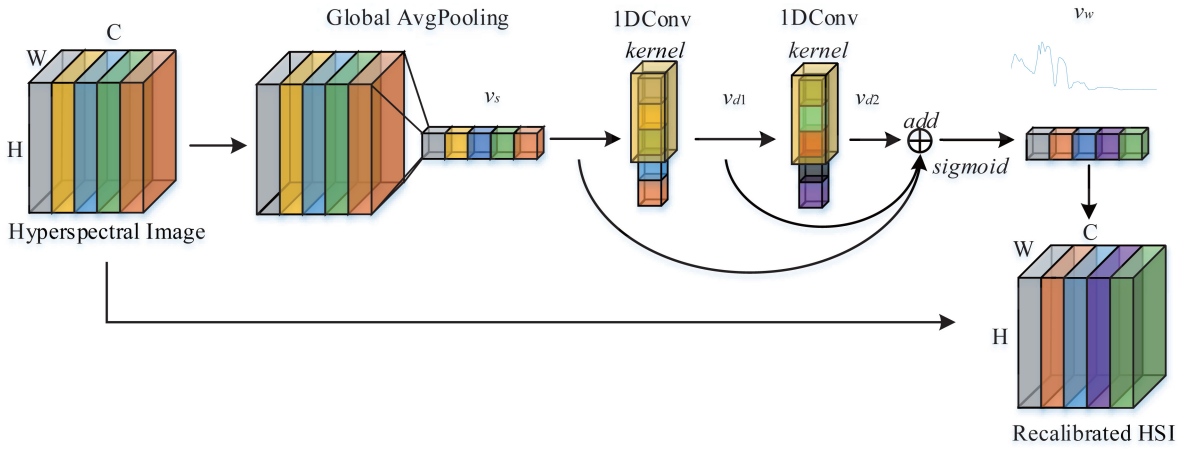


Fig. 2. Overall architecture of the proposed CBW module for HSIs classification. The CBW module is composed of the squeeze part, the capture part, and the aggregation part. The squeeze part is a global average pooling layer to obtain the global information to each band in the initial HSI patch. The capture part consists of two 1-D convolutional layers to capture the nonlinear correlations between each spectral band and its adjacent bands. The aggregation part aggregates the output of different layers by a fusion strategy to produce band weights, and the band weights are then used to recalibrate the HSI patch.

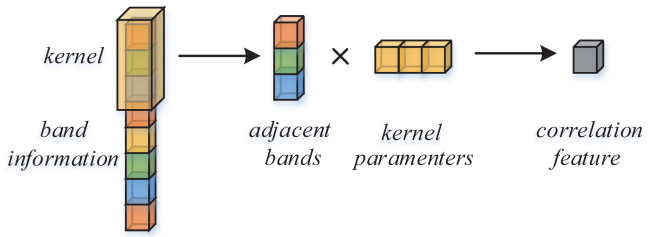


Fig. 3. Graphical representation of how to use 1-D convolutional layer achieve correlation feature from adjacent bands. First, several (the number equals the kernel size) adjacent band information picked by a 1-D convolution kernel. Then, matric multiplies the adjacent band vector and kernel parameters vector and then obtains adjacent bands correlation feature.

Thus, we applied the 1-D convolution of size K to capture the local dependencies v_d between K adjacent bands (neighbors) in the form

$$v_d^c = \sigma\left(\sum_{i=1}^{K} w_i v_s^i\right), \quad v_s^i \in \Omega_c^K \tag{3}$$

where v_d^c indicates the cth element of dependencies vector v_d, Ω_c^K represents the set of K-band statistics centered at v_s^i, w_i refers to the ith weight parameter of corresponding kernel, and $\sigma(\cdot)$ refers to the ReLU activation function. Simply, the dependencies vector v_d can be obtained by a 1-D convolutional layer, that is

$$v_d = \text{Conv1D}(v_s). \tag{4}$$

The 1-D convolution of different layers can quantify correlations of different orders, so two 1-D convolutional layers were built to capture local cross-band correlations. Each 1-D convolutional layer in the CBW module can obtain different order adjacent bands correlation features. However, it also lost some spectral statistics that are beneficial for classification. To address this problem in the CBW module, we used a fusion strategy to combine the spectral statistics with the correlations of different orders (termed shallow and deep) among the adjacent bands as follows.

Algorithm 1 CBW Algorithm

Input: C-band HSI patch $I = \{I_1, I_2, \ldots, I_C\} \in \mathbb{R}^{H\times W\times C}$;
Output: Recalibrated HSI $\widetilde{I} \in \mathbb{R}^{H\times W\times C}$;

1: Squeeze the global information of I patch into band-wise statistics vector v_s by Eq. (2);
2: Capture the band dependencies among adjacent bands v_{d1} (shallow) and v_{d2} (deep) by Eq. (3);
3: Aggregate v_{d1}, v_{d2} and v_s to generate band weight vector v_w by Eq. (5);
4: Recalibrate I by Eq. (6);
5: Reture $\widetilde{I}$.

In the final step, we aggregate the output of different layers to produce a band weight vector v_w to fully exploit the information from different layers, which can be expressed as

$$v_w = \delta(v_s \oplus v_{d1} \oplus v_{d2}) \tag{5}$$

where v_{d1} and v_{d2} refer to the dependencies vectors, which is the output vector of two cascaded 1-D convolutional layers, respectively [i.e., $v_{d1} = \text{Conv1D}(v_s)$ and $v_{d2} = \text{Conv1D}(v_{d1})$], $\oplus$ refers to bandwise addition, and $\delta(\cdot)$ refers to sigmoid function to obtain bands weight vector v_w that contains band weight values between 0 and 1. The band weight vector is then used to recalibrated the HSI patch by

$$\widetilde{I} = I \otimes v_w \tag{6}$$

where $\otimes$ denotes bandwise multiplication. The pseudocode of our CBW algorithm is shown in Algorithm 1.

B. CBW Module Design Analysis

The key point of the module design is to extract efficiently the correlation among the spectral bands by a lightweight structure.

To achieve this objective, a global average pooling layer with no adjustable parameters is included in CBW for squeezing input image into a vector that presents mainly the holistic spectral information.

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TABLE I

CONFIGURATION OF THE CBW MODULE AND THE SIMPLE CNN IN THIS ARTICLE. C REFERS TO THE NUMBER OF BANDS ON DATA SET, N-CLASS REFERS TO THE NUMBER OF CLASS ON DATA SET, AND BN REFERS TO THE BATCH NORMALIZATION LAYER

Model	Layer	Filters	Output shape	kernel	stride	padding	with
CBW	global avgpool	-	(1, 1, C)	11×11	-	-	-
	1Dconv1	1	(1, C)	7	1	3	BN + ReLU
	1Dconv2	1	(1, C)	7	1	1	BN + ReLU
	add	-	(1, C)	-	-	-	BN + sigmoid
2D-CNN	conv1	64	(11, 11, 64)	1×1	1	0	BN + Leaky ReLU
	maxpool	-	(6, 6, 64)	2×2	2	-	-
	conv2	128	(6, 6, 128)	3×3	1	1	BN + Leaky ReLU
	maxpool	-	(4, 4, 128)	2×2	2	-	-
	conv3	256	(4, 4, 256)	3×3	1	1	BN + Leaky ReLU
	maxpool	-	(3, 3, 256)	2×2	2	-	-
	fc1	-	(1024)	-	-	-	BN
	fc2	-	(N-class)	-	-	-	BN

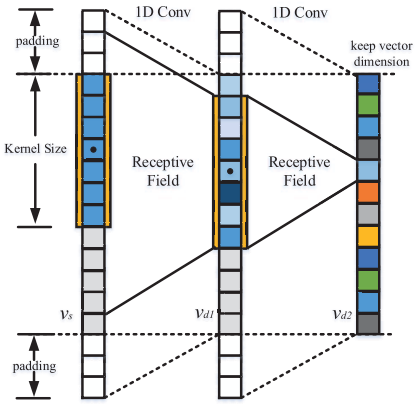


Fig. 4. Schematic on padding and receptive field (bands) in 1-D convolution operation.

Then, the capture component extracts the spectral local dependencies from adjacent bands by 1-D convolution. The number of receptive spectral bands is directly related to the kernel size K of the 1-D convolutional layer so that a larger kernel size can reveal the correlation of the wider bands range. Accordingly, padding of the convolutional layer will increase if the size of the output is to be kept as the same as the input. As shown in Fig. 4, the input vector is padded with $2 \times (K/2 + 1)$ zero elements for 1-D convolution operation to ensure the size of output unchanged from the input. In this case, some interference is introduced to the spectral dependencies, which affects the performance of the spectral weights. Thus, one cannot extend receptive spectral bands by increasing kernel size K. Another way to broaden the receptive spectral bands is to apply cascaded layers of 1-D convolution. Fig. 4 shows that the receptive spectral band of v_{d2} mapped from v_s can be enlarged by cascading 1-D convolution layers. The number of the receptive band can be increased to $K + L(K - 1)$ with L layers of 1-D convolution. By increasing the number of 1-D convolutional layers L, the band receptive spectral bands on v_s are enlarged. It means that cascading multiple layers of 1-D convolution can not only deepen the spectral semantic features but also expand the range of the receptive band without the detriment of interference from padding. Therefore, we applied two cascaded small-kernel 1-D convolution layers to capture the dependencies with different receptive spectral bands.

Finally, to make full use of the holistic spectral information and the spectral dependencies from adjacent bands, it is necessary to fuse them efficiently. We have developed the aggregation component of the CBW module to combine this spectral information by bandwise addition with no parameters.

Considering the tradeoff between lightweight and performance of CBW, the kernel size of 1-D convolution K and the number of 1-D convolutional layers L are set to 7 and 2, respectively, after extensive tests (more details are given in Section IV-F). After 1-D convolution, a batch normalization scheme is included in CBW that has two parameters. The total number of parameters of the proposed CBW module is given by $L \times (K + 2) + 2 = 20$.

C. Training Strategy

In this article, we utilize a single 2-D convolutional neural network (2D-CNN) as the backbone network for HSI classification. The structure of 2D-CNN is shown in Table I.

The optimizer and loss function of the CBW module are determined by backbone classification network configuration. In this work, the optimizer is Adam [43], which can efficiently update step size and can handle noise samples and effectively solve model parameters "cold start" problem (if initialized parameters are 0, the model needs to waste several items to eliminate the influence of "zero parameters" during start training). The cross-entropy loss is utilized as a loss function in this article, as follows:

$$\text{Loss} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{m=1}^{M}\left(p_n^m \cdot \log\left(q_n^m\right)\right) \tag{7}$$

where N is the number of samples in a minibatch, M refers to the number of class, and p and q is truth and prediction labels, respectively. The parameters in the CBW module are updated by backpropagation and optimizer of the backbone network.

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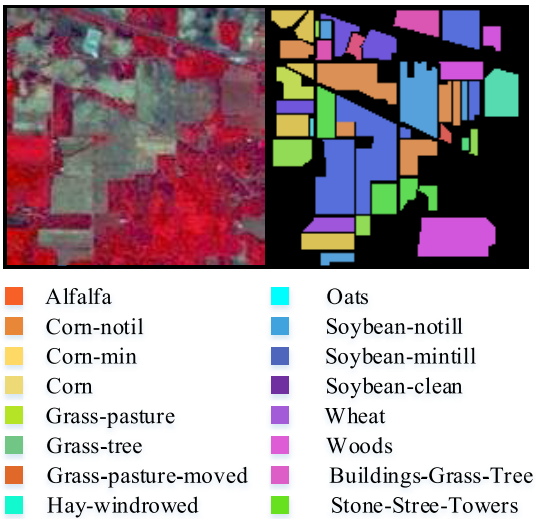


Fig. 5. AVIRIS Indian Pines data set. (Top) Three-band false-color composite and the ground-truth data. (Bottom) 16 class labels.

IV. Experimental Results

A. HSI Data Sets

The following data sets are used in our experiments: Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Indian Pines, Reflective Optics System Imaging Spectrometer (ROSIS) Pavia University image, and AVIRIS Salinas image:

1) The Indian Pines image set was acquired by the AVIRIS sensor over the agricultural Indian Pine test site in Northwestern Indiana in 1992. This scene has 200 bands across the wavelength range from 0.2 to 2.2 μm, and each band has a spatial pixel size of 145 $\times$ 145 pixels. The ground truth is divided into 16 classes. Before experiments, 20 water absorption bands (bands 104–108, 150–163, and 220) were removed. Fig. 5 shows the false-color composite of Indian Pines images and the corresponding ground-truth data.
2) The Pavia University image data set was recorded by the ROSIS sensor over an urban area surrounding the University of Pavia, Italy. The image is of size 610 $\times$ 340 $\times$ 103 and the wavelength range from 0.43 to 0.86 μm, with nine classes of interest. Fig. 6 shows a false-color composite of the Pavia University image and the corresponding ground-truth data.
3) The Salinas image set was also acquired by the AVIRIS sensor over the area of Salinas Valley, California. The image has 204 spectral bands of size 512 $\times$ 217, and the wavelength range is from 0.4 to 2.5 μm. Bands 108–112, 154–167, and 224 were removed due to water vapor absorption and low signal-to-noise ratio (SNR). This image set has 16 ground-truth classes, as shown in Fig. 7.

B. Experiment Setup

In order to quantitatively compare proposed methods with other different methods for HSIs classification tasks, the following measurements are considered.

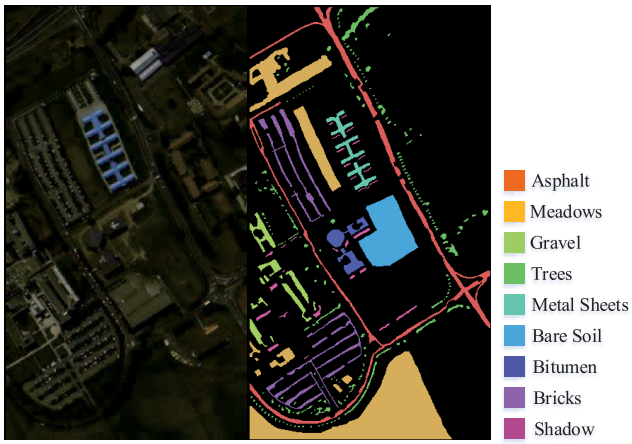


Fig. 6. ROSIS Pavia University data set. (Left) Three-band false-color composite. (Middle) Ground-truth data. (Right) Nine class labels.

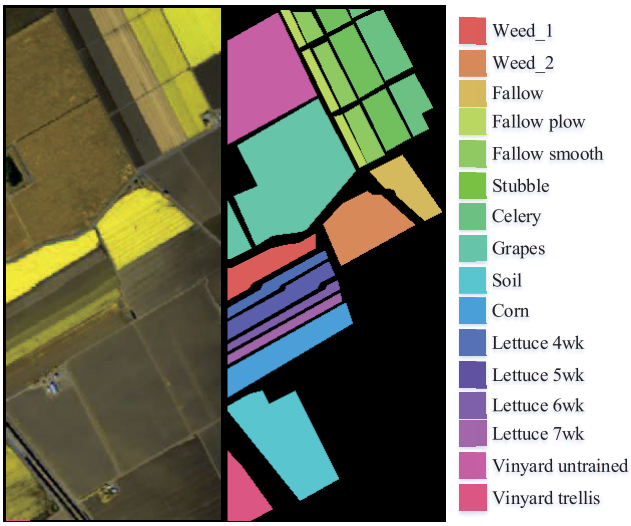


Fig. 7. AVIRIS Salinas data set. (Left) Three-band false-color composite. (Middle) Ground-truth data. (Right) 16 class labels.

1) *Overall Accuracy (OA):* OA means the ratio of the number of correctly classified category samples to the total number of categories. This criterion is calculated as the fraction of test samples that are differentiated correctly.
2) *Per-Class Accuracy:* To evaluate the effect of each category in the data set, we also calculated the accuracy of each category. This measurement is particularly useful when class labels are not uniformly distributed.
3) *Average Accuracy (AA):* This criterion is computed as the average of all per-class accuracies.
4) *Kappa:* This coefficient is an indicator used for consistency testing and can also be used to measure the effect of classification. In classification problems, the so-called consistency is whether the model prediction results are consistent with the actual classification results.

The followings are seven methods included in our contrast experiment.

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1) *SVM-RBF:* An SVM-RBF is a sum with radial basis function (RBF) kernel. We make use of fivefold cross-validation to search optimal hyperparameters γ (the inverse of the radius of influence of samples) and C (controlling the magnitude of penalization during the model optimization).
2) *2D-CNN:* To validate the effectiveness of the proposed method, a 2D-CNN is designed as a backbone network for classification by reference to the VGG network. As listed in Table I, the 2D-CNN consists of three convolutional layers and two full connectional layers. We utilize Leaky ReLU as an activation function and batch normalization after each layer.
3) *HybridSN [38]:* HybridSN is a spectral–spatial 3D-CNN followed by spatial 2D-CNN. Also, it is the state-of-the-art methods. In the HybridSN, the spectral dimension of HSIs is reduced to 30 by PCA. Three 3-D convolutional layers are set in the spectral–spatial learning module after PCA. Then, a 2-D convolutional layer, three full connection layers with dropout operation, and a softmax classifier are designed to achieve the classificational result. It is worth noting that the size of the HSI patch is set 25×25 to ensure functional completeness of HybridSN. Thus, we keep the patch size 25×25 for HybridSN in our experiment.
4) *specAtteNet [22]:* specAtteNet is a spectral attention module that makes an attention on solving HSIs spectral BW problem. It utilizes a 2-D global convolution to obtain band weights and adopts a gating mechanism to adaptively recalibrate spectral bands. Also, the 2D-CNN is used as the backbone network of specAtteNet. For simplicity, the 2D-CNN of specAtteNet embedding is denoted by specAtteNet + 2D-CNN. Comparison with specAtteNet can explore the differences of performance of BW module between 2-D global convolution and CBW.
5) *BW [14]:* BW is a band weighting module that designed and integrated an attention module embedded into traditional CNN for HSI classification. It has the capability to model the relationship among the bands and weight them according to their joint contribution to classification. One prominent feature of BW is that it can assign different weights to different pixel samples. Similarly, the 2D-CNN of BW embedding is denoted by BW + 2D-CNN.
6) *SE [18]:* The SE block is comprised of a convolutional layer, a global average pooling, and two FC layers that are connected to ReLU and sigmoid function, respectively. To demonstrate the superiority of the proposed method, a BW module is built based on an SE block that produces spectral weights with holistic spectral information. Similarly, the 2D-CNN of SE embedding is denoted by SE + 2D-CNN.
7) *BAM [23]:* BAM is a band attention module with the capacity of BW to implement the DL-based HSIs classification. BAM also can be seen as a plug-and-play complementary component of the existing classification networks when using CNNs for HSIs classification. The structure of BAM is more complicated than CBW. It is composed of five cascaded 2-D convolutional layers and two 1-D convolutional layers. Similarly, the 2D-CNN of BAM embedding is denoted by BAM + 2D-CNN.

In our experiment, training samples are generated by random sampling with class balance. For the Indian Pines scene, the number of training samples is 10% of the total samples. For Pavia University and Salinas scene, the number of training samples is only 1% of the reference data. The number of test samples is the total amount of samples excluding the training samples part. Our settings and hyperparameters are as follows: patch size $= 11 \times 11$ (HybridSN remain 25×25), batch size $= 100$, epoch $= 50$, and initial learning rate $= 0.001$. The learning rate is reduced by half after epoch 30. The Adam [43] algorithm is chosen as an optimizer and softmax as the classifier. We train network models on a PC with NVIDIA GTX 1060 3-GB GPU. To validate the robustness of the methods, every experiment was run repeatedly ten times, and the mean and variance of measurements (OA, AA, and kappa) were calculated as the final result.

C. Comparative Assessment

First, an ablation experiment comparing CBW + 2D-CNN against 2D-CNN is performed on three data sets to validate and evaluate the effectiveness of the proposed CBW module. The results are summarized in Tables II–IV. Specifically, we can observe that CBW + 2D-CNN outperforms the 2D-CNN by: 1) 1.20% of OA, 1.24% of AA, and 1.59 of Kappa coefficient on the Indian Pines data set; 2) 1.26% of OA, 3.71% of AA, and 1.77 of Kappa coefficient on the Pavia University data set; and 3) 2.27% of OA, 0.78% of AA, and 1.86 of Kappa coefficient on the Salinas data set. CBW can reduce accuracy fluctuation (see the variance within the braces in Tables II–IV). The results prove that our proposed CBW module effectively enhances the representation ability of band information and improves the classification effect and robustness of the backbone network.

On the other hand, in order to further explore the effect of CBW for HSI classification, some HSI classification methods are applied to compare with CBW, including SVM-RBF, HybridSN [38], specAtteNet [22], BW [14], SE [18], and BAM [23]. These methods, except SVM-RBF, are DL-based classification methods. As DL-based methods are capable of strong feature learning, usually, they show better performance than traditional classification methods.

For DL-based methods, the number of training samples is a very important factor. HybridSN is one of the state-of-the-art end-to-end DL-based method for HSI classification [38]. However, HybridSN requires sufficient samples to ensure its excellent performance. In our experiment, a challenging training condition is considered, and 10% data and 1% instead of both 30% in [38] data are selected as training set for Indian Pines and Salinas, respectively. As presented in Tables II and IV, the performance of 2D-CNN is not good as HybridSN, but the performance of CBW + 2D-CNN

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TABLE II

CLASSIFICATION ACCURACY OF THE SVM-RBF, 2D-CNN, HYBRIDSN, 2D-CNN WITH SPECATTENET, BW, SE, BAM, AND CBW MODULES IN THE INDIAN PINES SCENE, RESPECTIVELY. THE VARIANCE OF VARIOUS MEASUREMENT IS NOTED IN BRACKETS

The number of training samples is 10% of the reference data.								
Class	**SVM-RBF**	**2D-CNN**	**HybridSN**	**2D-CNN**				
				specAtteNet	**BW**	**SE**	**BAM**	**CBW**
Alfalfa	68.92	97.83	97.83	97.78	**99.25**	96.63	97.76	97.84
Corn_N	77.92	91.20	94.99	94.78	94.17	91.96	95.11	**95.33**
Corn_M	79.04	94.67	95.34	96.79	94.47	97.02	**97.33**	96.56
Corn	67.92	90.66	95.84	93.99	98.54	98.11	**98.16**	96.73
Grass_P	88.65	96.80	96.16	97.43	**99.06**	98.54	97.23	98.10
Grass_T	89.13	98.58	98.42	96.50	97.14	98.84	98.52	**98.92**
Grass_M	95.43	98.18	96.55	92.59	97.63	96.30	98.11	**98.25**
Hay_W	97.34	99.89	99.58	99.90	99.23	99.27	99.48	**99.92**
Oats	57.25	93.44	81.82	72.73	**94.74**	91.89	92.27	91.89
Soybean_N	78.75	96.15	95.02	95.37	97.86	92.63	**97.77**	96.00
Soybean_M	81.13	97.23	96.78	97.20	98.75	96.96	**98.84**	97.34
Soybean_C	77.75	91.71	92.54	94.64	93.74	93.57	**96.28**	94.63
Wheat	93.69	98.32	98.32	98.74	99.51	**100**	**100**	**100**
Woods	92.79	98.67	99.09	97.85	99.92	97.87	96.36	**99.89**
Buildings	74.45	94.72	**98.84**	94.04	93.66	93.26	96.28	97.75
Stone	**99.15**	93.47	94.05	98.40	95.56	92.76	96.26	96.74
OA	83.09(1.86)	95.85(2.10)	97.01(2.35)	96.47(1.32)	96.41(1.22)	96.41(4.15)	96.83(1.16)	**97.05(0.84)**
AA	82.46(1.76)	95.97(2.33)	95.67(2.22)	94.92(1.72)	97.07(1.48)	95.98(3.85)	97.20(1.57)	**97.21(0.79)**
Kappa	80.65(1.64)	95.30(1.98)	96.86(2.01)	96.13(1.22)	96.02(1.17)	95.44(3.65)	96.59(1.33)	**96.89(0.71)**

TABLE III

CLASSIFICATION ACCURACY OF THE SVM-RBF, 2D-CNN, HYBRIDSN, 2D-CNN WITH SPECATTENET, BW, SE, BAM, AND CBW MODULES IN THE PAVIA UNIVERSITY SCENE, RESPECTIVELY. THE VARIANCE OF VARIOUS MEASUREMENT IS NOTED IN BRACKETS

The number of training samples is 1% of the reference data.								
Class	**SVM-RBF**	**2D-CNN**	**HybridSN**	**2D-CNN**				
				specAtteNet	**BW**	**SE**	**BAM**	**CBW**
Asphalt	85.51	93.90	85.08	94.62	93.90	**95.91**	95.23	95.32
Meadows	93.05	98.95	98.04	97.47	98.97	98.80	**98.69**	**99.05**
Gravel	73.52	**73.95**	71.65	81.38	81.30	84.11	86.41	**86.84**
Trees	91.93	98.17	95.32	98.00	98.26	**98.47**	98.27	98.44
PaintedM_P	99.06	97.12	99.41	99.63	99.67	99.52	**99.96**	99.88
Bare_S	77.46	99.20	96.37	93.72	99.12	98.36	98.19	**99.84**
Bitumen	72.07	79.94	71.50	82.55	76.38	85.78	85.17	**88.88**
Bricks	83.49	85.41	83.94	86.54	87.36	87.91	90.86	**90.92**
Shadows	**99.87**	95.79	84.76	96.72	97.67	95.85	98.19	96.60
OA	87.77(0.85)	94.57(1.07)	91.68(1.12)	94.67(1.21)	95.60(0.36)	95.44(0.62)	95.76(0.36)	**95.83(0.31)**
AA	86.22(0.82)	91.38(1.87)	87.34(0.93)	92.29(1.53)	92.51(1.24)	93.86(1.48)	94.55(1.08)	**95.09(0.93)**
Kappa	83.82(0.76)	93.00(0.98)	88.98(0.99)	93.13(0.87)	93.37(0.40)	94.48(0.47)	94.66(0.34)	**94.77(0.27)**

outperforms HybridSN. The results show that our proposed CBW is able to help the DL-based model overcome the poor training conditions.

Other methods, specAtteNet, BW, SE, and BAM, are the same as CBW, in essence, belongs to the DL-based BW module. Compared with specAtteNet, CBW achieves 0.58%, 2.29%, and 0.76 gain in terms of OA, AA, and Kappa coefficient, respectively, on the Indian Pines data set. Regarding the Pavia University and Salinas data sets, CBW achieves 1.16%, 2.80%, and 1.64 and 1.85%, 0.75%, and 2.17 gains in terms of OA, AA, and Kappa coefficient, respectively. Meanwhile, the lower variance of measurements denotes that the CBW module is more stabilized and robust than specAtteNet. Similarly, the performance of CBW is better than other BW modules, such as BW, SE, and BAM. The classification maps of all experiment methods on three data sets are shown in Figs. 8–10. For the Indian Pines data set shown in Fig. 8, sample misclassifications occur less in the classification map by the CBW + 2D-CNN method than other maps, and the misclassifications locate mainly near edges of different classes. Similarly, for the PaviaU data set shown in Fig. 9, the CBW + 2D-CNN can discriminate the Bricks class samples from Gravel class samples more accurately than other classification methods. For the case of the Salinas data set in Fig. 10, one can see clearly that the samples of the Grapes class and the Vinyard untrained class are often mislabeled by other methods, but the CBW + 2D-CNNmethod performs favorably in this regard with significantly reduced errors. The comparisons show that CBW is able to capture the important band information of the HSIs more accurately and efficiently for classification, and CBW can also enhance the performance and robustness of 2D-CNN.

D. Efficiency and Complexity Analysis

We have performed additional experiments using CPU only of a board-level minicomputer (Raspberry Pi 4B). As shown

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TABLE IV

CLASSIFICATION ACCURACY OF THE SVM-RBF, 2D-CNN, HYBRIDSN, 2D-CNN WITH SPECATTENET, BW, SE, BAM, AND CBW MODULES IN THE SALINAS SCENE, RESPECTIVELY. THE VARIANCE OF VARIOUS MEASUREMENT IS NOTED IN BRACKETS

The number of training samples is 1% of the reference data.								
Class	**SVM-RBF**	**2D-CNN**	**HybridSN**	**2D-CNN**				
				specAtteNet	**BW**	**SE**	**BAM**	**CBW**
Brocoli_G1	98.52	98.38	99.75	99.64	99.36	97.24	99.97	**99.98**
Brocoli_G2	99.08	99.07	99.83	99.74	99.67	98.45	99.87	**99.88**
Fallow	91.43	97.13	99.14	97.20	91.67	96.55	98.28	**99.45**
Fallow_R	96.22	97.86	96.04	98.46	98.40	99.68	**99.86**	96.28
Fallow_S	97.38	96.73	98.73	99.46	96.08	98.83	**98.79**	97.81
Stubble	99.58	99.86	97.06	99.85	99.92	99.92	99.95	**99.97**
Celery	99.24	99.36	99.68	99.65	**99.97**	99.82	99.93	98.11
Grapes_U	72.41	88.20	87.29	85.00	88.66	89.69	88.29	**90.99**
Soil_V	97.78	99.86	99.64	**99.89**	99.48	95.15	99.82	99.59
Corn_S	87.59	98.00	**98.83**	96.93	96.85	97.74	97.63	98.03
Lettuce_R4	82.41	99.25	**99.39**	95.78	96.82	95.01	97.00	97.53
Lettuce_R5	98.47	99.95	99.43	99.95	98.16	99.27	**99.93**	99.82
Lettuce_R6	95.81	98.97	97.03	99.89	97.77	99.95	98.34	**99.98**
Lettuce_R7	94.38	98.64	98.39	97.22	98.33	98.91	98.66	**99.35**
Vinyard_U	66.43	78.49	82.93	81.94	81.00	86.39	78.18	**87.80**
Vinyard_V	96.81	98.51	99.16	98.15	99.39	98.32	**99.83**	96.22
OA	87.27(1.21)	93.55(1.30)	94.35(2.45)	93.97(2.39)	94.42(4.24)	94.03(3.88)	95.49(1.26)	**95.82(0.31)**
AA	92.10(0.98)	96.77(0.89)	97.02(1.33)	96.80(1.26)	96.34(2.35)	96.93(1.56)	97.10(0.59)	**97.55(0.26)**
Kappa	85.93(1.07)	93.41(1.26)	93.72(2.32)	93.10(2.11)	93.47(3.89)	93.32(3.42)	94.75(1.16)	**95.27(0.30)**

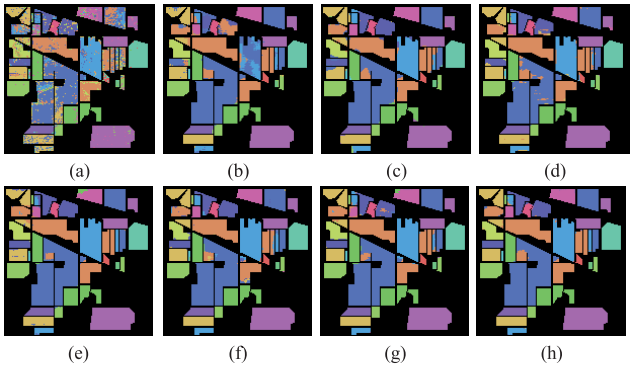


Fig. 8. Classification maps of different methods for Indian Pines data set. (a) SVM-RBF. (b) 2D-CNN. (c) HybridSN. (d) specAtteNet + 2D-CNN. (e) BW + 2D-CNN. (f) SE + 2D-CNN. (g) BAM + 2D-CNN. (h) CBW + 2D-CNN.

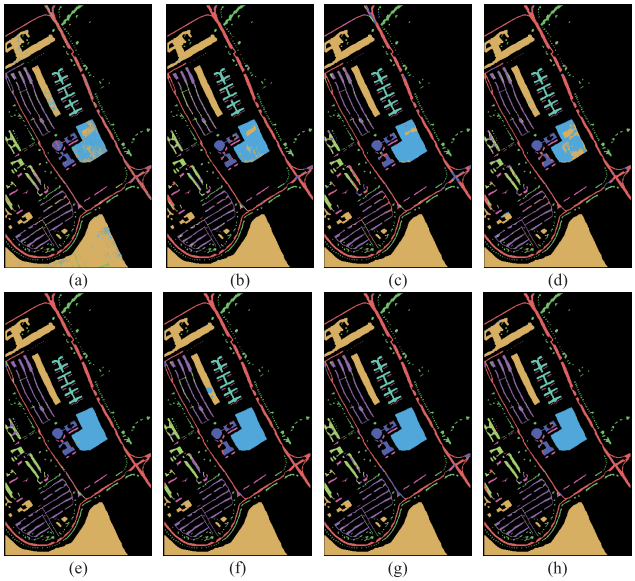


Fig. 9. Classification maps of different methods for Pavia University data set. (a) SVM-RBF. (b) 2D-CNN. (c) HybridSN. (d) specAtteNet + 2D-CNN. (e) BW + 2D-CNN. (f) SE + 2D-CNN. (g) BAM + 2D-CNN. (h) CBW + 2D-CNN.

in Fig. 11, the computing power and cost of chosen computer are far lower than a high-performance computer with a dedicated GPU board for running deep network codes but provide an easy-to-implement approach for many field applications.

Generally, the numbers of module parameters and floating-point operations per second (FLOPs) are used as metrics to evaluate the model computational complexity. Table V lists these metrics for evaluation of different classification methods on three data sets. In the case of the Indian Pines data set, the values of FLOPs of specAtteNet, BW, SE, and BAM module are 4.83M, 6.05M, 4.94M, and 3.92M, and the numbers of parameters are 4 840 200, 4 876 300, 40 900, and 108 099, respectively. In comparison, the CBW module needs merely 20 parameters and 0.03M FLOPs, which demonstrates clearly that the proposed method has a much reduced model and computational burden. Moreover, different from other methods, the number of parameters of CBW does not change with the dimension and size of input HSIs. This characteristic ensures that CBW is lightweight and robust for most case (for HSIs with high spectral dimensionality and spatial resolution). Furthermore, the lightweight feature of CBW can decrease significantly memory requirement for field applications.

We have also analyzed the extra time cost associated with a BW module embedded into a backbone network, such as 2D-CNN. For this purpose, the extra time and relative percentage are defined with the latter given by the ratio of

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TABLE V

NUMBER OF MODEL PARAMETERS AND FLOPS OF DIFFERENT DL-BASED BW METHODS ON INDIAN PINES (IP), PAVIA UNIVERSITY (PU), AND SALINAS DATA SETS, RESPECTIVELY

Model	FLOPs			#Parameter		
	IP	PU	Salinas	IP	PU	Salinas
2D-CNN	11.36M	10.61M	11.39M	2,761,584	2,755,376	2,761,840
specAtteNet	4.83M	1.31M	5.04M	4,840,200	1,283,792	5,035,740
BW	6.05M	2.42M	6.17M	4,876,300	1,314,907	5,072,562
SE	4.94M	1.32M	5.13M	82,127	15,727	85,381
BAM	3.92M	2.23M	3.99M	108,099	61,139	110,192
CBW	**0.03M**	**0.01M**	**0.03M**	**20**	**20**	**20**

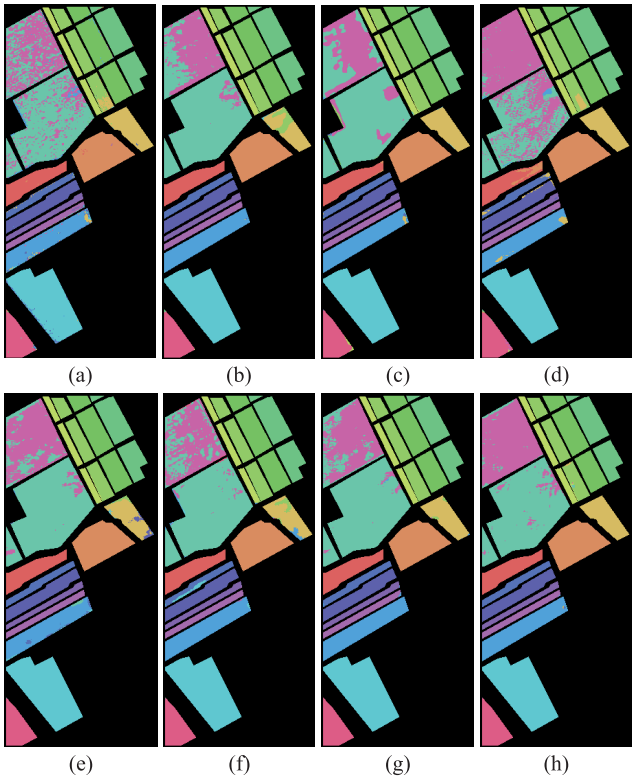


Fig. 10. Classification maps of different methods for Salinas data set. (a) SVM-RBF. (b) 2D-CNN. (c) HybridSN. (d) specAtteNet + 2D-CNN. (e) BW + 2D-CNN. (f) SE + 2D-CNN. (g) BAM + 2D-CNN. (h) CBW + 2D-CNN.



Fig. 11. Physical image of Raspberry Pi 4B (RPi 4B).

the extra time spent in the BW module to the time of the stand-alone 2D-CNN to perform identical classification tasks. Table VI presents the results of time analysis for different

TABLE VI

RUNNING TIME OF BW METHODS ON RPI 4B

Methods	IndianPines(200)	PaviaU(103)	Salinas(204)
2D-CNN	106s	966s	550s
BW	+125s(+117%)	+588s(+61%)	+3316s(+602%)
SE	+180s(+170%)	+935s(+97%)	+1222s(+222%)
speAtteNet	+78s(+73%)	+215s(+22%)	+340(+62%)
BAM	+69s(+65%)	+229s(+24%)	+138s(+25%)
CBW	**+16s(+15%)**	**+139s(+14%)**	**+56s(+10%)**

BW modules on the three data sets. As shown in Table III, 2D-CNN took 106, 966, and 550 s to perform classification tasks on the three data sets respectively. Using these times as the baseline values, the BW module exhibits markedly increased time cost due to its high model complexity, which even grows to 3316 s or about sixfold longer than that of 2D-CNNs on the Salinas data set. The SE block took less extra time for convolution calculation in the case of Salinas with large input HSI spectral bands. However, it requires a much longer time for high-dimension matrix calculation of the fully connected layer. The BAM and specAtteNet require relatively smaller extra time costs but still cost more than 20% on the Pavia University data set, more than 65% on the Indian Pines data set of the baseline values. In contrast, our CBW module took only 16 s in extra time or 15% of the baseline value, 139 s or 14%, and 56 s or 10% to complete the task. The above experimental results show that the lightweight feature of the CBW module allows not only the much reduced need for storage space but also the significant increase in computing speed.

E. BW Analysis

In order to evaluate the recalibrated effect of CBW module, a parameter S_m is used to quantify intraclass similarity. The intraclass similarity measure is defined as follows:

$$S_m = \sum_{c=1}^{C} \sum_{i=1}^{N_c} (x_{i,c} - \mu_c)(x_{i,c} - \mu_c)^T \tag{8}$$

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} x_{i,c} \tag{9}$$

where $x_{i,c}$ refers to the ith sample of class c. N_c denotes the number of samples of class c, and μ_c is average spectral vector of class c. The greater S_m value, the more dispersed for data within the same class. Conversely, lower S_m value means

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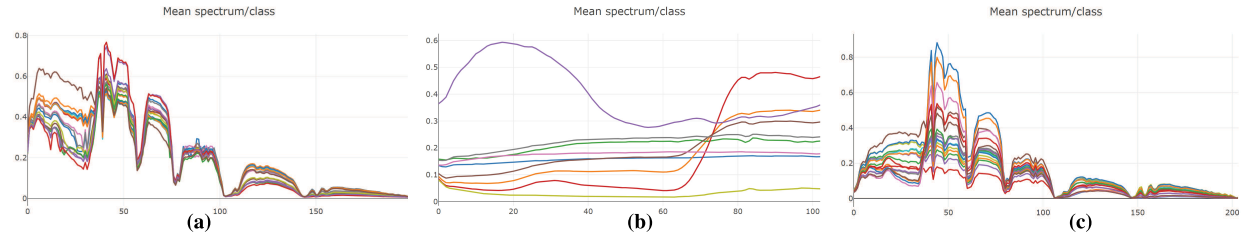


Fig. 12. Average spectral plots from each data set. (a) Indian Pines. (b) Pavia University. (c) Salinas. Each curve represents the average distribution of the spectra for a particular class in the data set. The Indian Pines, Pavia University, and Salinas data sets have 16, 9, and 16 class labels, respectively.

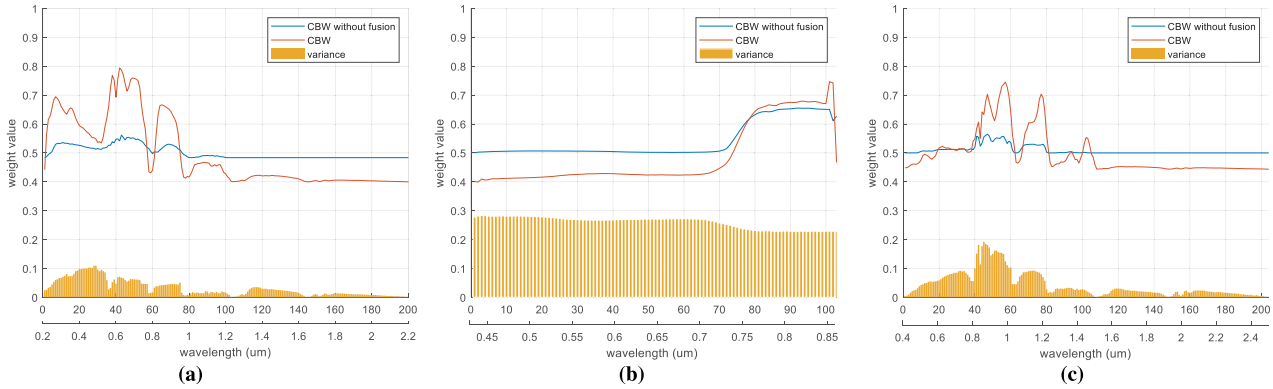


Fig. 13. BW values distribution of CBW module with fusion strategy and without fusion strategy for (a) Indian Pines, (b) Pavia University, and (c) Salinas data sets. The variance of intraclass average spectra is displayed for assisting analyze.

TABLE VII

WITHIN-CLASS SIMILARITY MEASURE BEFORE CBW AND AFTER CBW ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATA SETS

Dataset	Before($\times 10^3$)	After($\times 10^3$)
Indian Pines	1.8892	**1.5166**
Pavia University	9.9188	**5.7068**
Salinas	1.7702	**1.0352**

TABLE VIII

COMPARATIVE EXPERIMENT RESULTS OF CBW WITH FUSION STRATEGY AND WITHOUT FUSION STRATEGY ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATA SETS

Methods	IndianPines(10%)	PaviaU(1%)	Salinas(1%)
CBW without fusion	96.64%	95.63%	94.82%
CBW with fusion	97.05%	95.80%	95.82%

high intraclass similarity. Generally, data with high intraclass similarity are easier to be separated into different classes for their tighter intraclass aggregation. As shown in Table VII, the S_m values decrease after three HSI data sets are recalibrated by the proposed CBW module. This means that the application of CBW module enhances the intraclass similarity and, thus, improves the classification performance.

To further understand the relative importance of the fusion strategy for CBW, we conduct an empirical compassion experiment on the application of the CBW with and without fusion strategy. Specifically, when fusion strategy is applied in the final aggregation step, we have $\upsilon_w = \delta(\upsilon_s \oplus \upsilon_{d1} \oplus \upsilon_{d2})$, as given in (6). Otherwise, it becomes $\upsilon_w = \delta(\upsilon_{d2})$ without using the fusion strategy. Fig. 12 shows the arithmetically averaged spectral signatures of each band in each class on the Indian Pines, Pavia University, and Salinas data sets. The total number of bands for the three data sets is 200, 103, and 204, respectively. The band weights are displayed for each considered data set, and the red and blue plots correspond to the band weights obtained by CBW with fusion strategy and without fusion strategy, respectively. As shown in Fig. 13, we use the yellow bars to represent the variance of intraclass average spectral. As illustrated in Figs. 12 and 13, compared with CBW without fusion strategy, the band weight distribution becomes more matched with the distribution of the variance of intraclass average spectral by adopting fusion strategy in CBW. In other words, the band weight obtained by CBW with fusion strategy can more accurately enhance high-contribution bands and suppresses low-contribution bands and noise. As shown in Table VIII, the performance of the CBW module with fusion strategy improves significantly over the module without fusion strategy on the three data sets. Hence, the fusion strategy in the aggregation step plays an important role in CBW.

F. Parameter Analysis

To investigate the optimal kernel size of 1-D convolutional layers for CBW, different settings have been tested in a range of 3–199 on Indian Pines and Salinas and 3–101 on Pavia University. Fig. 14(a) displays the performance of BW for increasing kernel size K on three data sets, which increases to a peak value and then decreases. These results show

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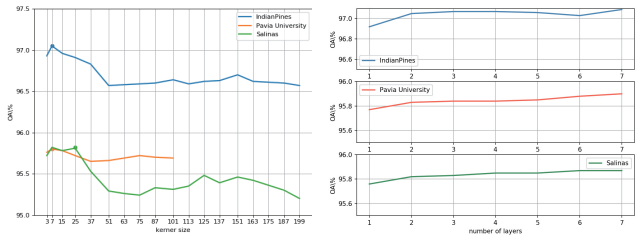


Fig. 14. Experimental result of (a) different kernel sizes and (b) different numbers of 1-D convolutional layers in CBW on Indian Pines, Pavia University, and Salinas data sets. The point refers to the highest OA on the corresponding data set.

empirically that excessive kernel sizes lead to performance deterioration. When the kernel size is set to 7, CBW achieves the best performance on Indian Pines and Salinas data sets and works well on the Pavia University data set. Thus, we set the kernel size of 1-D convolution to 7 for our CBW module.

The number of cascading layers for 1-D convolution L is also an important parameter that decides the depth of spectral feature extraction and the receptive band related to local dependencies. As seen from Fig. 14(b), increasing L does not raise the performance of CBW much, and two 1-D convolutional layers are adopted in CBW to balance performance and efficiency.

V. Conclusion

In this article, we focused on utilizing effective band weights for learning their correlation patterns and improved DL-based HSIs classification with low computational complexity. To this aim, we developed a new module architecture termed CBW on the basis of the attention mechanism. As a notable characteristic, the proposed CBW module has merely 20 model parameters that are far less than those used in the other existing modules and enables implementation in compact devices. Extensive experiments demonstrate the effectiveness of the CBW module, and the 2D-CNN with CBW embedding achieves state-of-the-art performance on three widely used HSI data sets. The results further show that the better performance of CBW is achieved with a lower computational burden in comparison to other BW modules.

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